



OPEN Multi-output deep learning for high-frequency prediction of air and surface temperature in Kuwait

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Accurate prediction of air and surface temperature is essential for urban planning and climate resilience, especially in arid regions. This study evaluates the performance of multi-output regression models using high-frequency climate data collected every 5 min over four years in Kuwait. Thirty environmental variables (e.g., including humidity, solar radiation, dew point, and wind direction) were used to predict six air and surface temperature-related outcomes simultaneously. Ten models, including deep learning and traditional machine learning approaches, were benchmarked using a leave-1-year-out validation strategy. Results show that contextual embeddings-based Transformer (FTTransformer) and Long Short-Term Memory (LSTM) achieved strong predictive performance with an R^2 of 0.998, a mean squared error of 0.13, and a mean absolute error of 0.24 when forecasting six temperature variables at 5-min resolution. These results significantly outperform traditional machine learning models and demonstrate the robustness of deep learning approaches for high-frequency climate prediction. While deep learning models outperformed conventional methods, LSTM's performance degraded on anomalous data from previous years, whereas FTTransformer maintained stable accuracy across years. Model interpretation using SHAP and permutation importance identified key predictors for this task, underlining the significance of diverse climate features.

Keywords Regression, Multioutput, Climate, Machine learning, Deep learning

Kuwait is a country in West Asia, situated on the northern edge of the Arabian Peninsula at the head of the Persian Gulf. The country experiences some of the harshest summer temperature in the broader Middle East region; and is undergoing climate and temperature changes since the known records from 1960s^{1,2}. It is estimated that by the mid of current century, the average temperature in Kuwait could increase by 1.80°C – 2.57°C ³. At large, these changes can significantly affect daily activities⁴, human health⁵, economic prosperity⁶, heat-related mortality³, degrading biodiversity⁷, disrupting ecology⁸, threatening terrestrial ecosystems, including wildlife habitat⁹ and worsening agricultural outcomes¹⁰. These impacts are along the lines of the larger geographical region of the Middle East^{11,12}.

Climate change is a gradual phenomenon and often necessitates decades of data to understand the slow moving changes in the temperature and other elements. However, short-term climate data may also yield insights into current weather patterns. It is known that several key climatic factors, such as wind speed, relative humidity, radiation, etc., can be utilized to predict air and surface temperatures. However, their application for long-term predictions or annual forecasting is inadequately studied. Given multidimensional climate data, one of the key questions can be whether this data from the previous years is useful to predict future year's air and surface temperatures? If such an accurate predictive model is built, this can be used to measure any major deviations in a given year, and the reasons for climate change could be analyzed. Advancements made in artificial intelligence and machine learning (ML) have enabled to process, analyze and utilize vast amounts of climate data to achieve both short-term and long-term prediction abilities^{13–15}.

Temperature prediction models have been developed for several geographical regions, which can predict average, maximum, or minimum temperatures at different time horizons (1–24 h)^{16–20}. There is also some research for predicting longer horizon temperature, such as one year ahead²¹ or next season^{22,23}. Predicting long term temperature trends can provide valuable insights into current climate changes; however, it entails specific advantages and limitations. Long-term prediction of temperature may assist in strategic planning for climate sensitive sectors (e.g., agriculture), and disaster preparedness (e.g., extreme temperatures)²⁴. On the other hand, most of the proposed methods are probabilistic may inadequately address climatic uncertainties, and therefore may provide reduced accuracy^{13,24}. These models may suffer from coverage bias, where incomplete training data

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used to train the models can lead to an underestimation of climatic variables²⁵. The prediction task could become more challenging if more than one type of outcomes is to be predicted at specific times in a longer horizon.

Recent advances in deep learning (DL) have shown promising results in climate and hydrological forecasting tasks, such as rainfall and precipitation prediction^{26–28}. Some studies use DL approaches to predict other environmental variables, e.g. global sea level²⁹ and methane concentrations³⁰ for a longer horizon in future (up to the year 2050). These studies demonstrate the potential of DL approaches in handling high-resolution climate time series data for both shorter- and longer-term time horizon. However, these prior efforts have focused on either tropical or global environments, whereas our study uniquely addresses high-frequency, multi-output temperature prediction in an arid environment (Kuwait) using different types of DL methods. The goal of this paper is to develop DL models to establish the inter-annual robustness of high frequency multidimensional climate data to predict multiple air and surface temperature variables. By adopting a leave-one-year-out evaluation strategy, we explicitly test model generalization across distinct seasonal cycles, which is particularly important for practical applications, such as urban planning, ecotourism and climate resilience in the region.

Numerous studies have been conducted to forecast temperatures at different time horizons in the broader Middle East region^{31–33}. However, to the best of our knowledge, there are no Kuwait specific temperature prediction studies published so far. This study presents the development and evaluation of ten advanced ML, deep learning (DL) and sequential DL algorithms to predict six outcomes related to air and surface temperatures at a longer horizon of one-year. We employed explainability methods—SHAP and permutation importance—for temperature modeling, which are rarely used for multivariate weather data. While Transformer-based architectures have been applied in weather and climate prediction, they have primarily focused on temporal sequence modeling^{34–36}. In contrast, this study employs the contextual embeddings-based Transformer (FTTransformer), which is specifically designed for heterogeneous tabular data with multiple correlated features. The architecture introduces contextual embeddings and feature-wise self-attention, enabling the model to capture nonlinear interactions across diverse climate indicators, such as humidity, solar radiation, and air density. This design allows FTTransformer to provide both strong predictive accuracy and interpretability. To the best of our knowledge, this is the first application of FTTransformer to high-frequency (5-min resolution) climate data for multi-output temperature prediction in arid regions. Our particular contributions are as follows:

- We analyzed four years worth of climate data collected on site at the American University of the Middle East, Kuwait from 2021 to 2024. The data is collected at a high resolution of every five minutes.
- We formulated the temperature prediction problem as a multi-output regression problem. The input includes 30 climatic variables capable of simultaneously predicting six outcomes—surface temperature and air temperature, and their maximum and minimum values.
- The results showed that the FTTransformer and Long short Term Memory (LSTM) achieved the best average R^2 score, MSE and MAE over four years (approximately 0.998, 0.13 and 0.24, respectively). Monthly predictions of temperature show the robust performance of FTTransformer in comparison to LSTM
- SHAP and permutation importance provided both local and global insights into the significance and usefulness of several climate variables, identifying dew point and relative humidity as the primary predictors.

Related work

Weather forecasting largely involves predicting and describing short-term atmospheric conditions, while climate modeling focuses on long-term trends and changes in atmospheric factors. Although Numerical Weather Prediction methods are the benchmark, this type of method faces challenges such as inherent atmospheric uncertainty and computational cost. The advent of ML and, more recently, the emergence of DL have transformed the field through data-driven models¹⁵. In this section, we will discuss climate studies that use ML and DL approaches for long-term prediction of temperature. Short-term prediction and other methods are available in the existing reviews^{37,38}.

Cachay et al.³⁹ presented a dynamic informed diffusion model for multi-step forecasting and long-range probabilistic spatiotemporal forecasting. They tailor diffusion models to support long range forecasting by coupling the diffusion process and the dynamical nature of the data. Their method imposes an inductive bias and improves computational efficiency compared to Gaussian noise-based diffusion models. Their task of is to forecast a longer horizon from a single initial condition. For a sea surface dataset, they train, validate, and test the model for the years 1982–2019, 2020, and 2021, respectively. Fister et al.²² presented several DL, ML and data reduction techniques to predict long-term summer air temperature. They aim to predict the first and second fortnights for the month of August using input data from previous months, at two different locations in France and Spain. Their results show that the data reduction technique of recurrence plots when combined with a convolutional neural network (CNN) performs best for both shorter and longer horizons in comparison to other studies baselines. This technique was able to accurately detect some maximums in the summer temperature better than other techniques. Pérez-Aracil et al.²³ presented two autoencoder based models for long-term temperature prediction across various European cities that encountered major heatwaves. Their approach comprises using an autoencoder to detect anomalies between inputs and outputs, which are then input into the second-stage model that may be either another autoencoder or a multilayer neural network. They used the hourly data from 1950 to 2022. Their goal was to predict 2-m temperature based on seven other climate variables up to 4 weeks in advance. Tran et al.⁴⁰ employed a genetic optimization algorithm along with several neural network architectures to first select the best architecture, and then predict the 15-day-ahead daily maximum temperature in the summer. The LSTM model optimized by the genetic algorithm gave better results compared to other models. The data were collected at the Cheongju station in South Korea, covering the period from 1976 to 2015.

Nandi et al.⁴¹ present a transformer based approach to predict temperature up to 150 days ahead with high accuracy. They trained their models with more than 25 days of data from 5 cities around the globe. They added

a time encoding vector to learn temporal ordering and a convolution block to learn the seasonal aspect of the data. Their model outperformed other ML and DL baselines. However, this approach is limited to univariate time series datasets. Durhasan et al.⁴² applied statistical and sequential DL models to predict the monthly mean surface temperatures of future years. They trained the models from January 1880 to December 2008, and tested the models from January 2009 to December 2022. They further develop prediction models until 2050, which show a rise in global temperature of up to 1 °C.

Ayad⁴³ presented the use of SARIMA model to predict monthly air temperature from the Sania Ramel station, Morocco. The SARIMA model was estimated and validated using the data over the period from January 1980 to July 2022, and used to predict the average air temperature from August 2022 to January 2024. However, no other model comparisons were shown in this work. Shrivastava et al.⁴⁴ used multivariate polynomial regression and deep neural network models to predict the next year's temperature in 6 hour intervals. They trained the model on the past 6 years of data. Park et al.⁴⁵ first refined missing data in a weather dataset and achieved higher performance with an LSTM model in comparison to a deep network over a time horizon of up to 14 days. Their data came from the Meteorological Office of South Korea comprising hourly measurements for temperature, wind direction, wind speed, relative humidity, cumulative precipitation, vapor pressure, and barometric pressure. They trained the models on the data from November 1981 to December 2016 and tested them on a year's data from January 2017 to December 2017. Roy⁴⁶ used the weather data collected at the John F. Kennedy International Airport, USA to predict the temperature for 1 and 10 days in the future. The input to the model was the data records related to the past seven days. Their results showed that a combination of a CNN and LSTM outperformed other models.

The above discussion indicates that sequential DL models, such as LSTM and transformers, can be good at predicting long term air temperature. However, under certain circumstances, CNN, Recurrent Neural Networks (their combination) and other classical conventional ML models may also be effective. However, RNN models encounter constraints in long-term air temperature forecasting due to their challenges in capturing long-range relationships, mainly attributed to vanishing or exploding gradients during the training process⁴⁷. This problem limits their capacity to retain and employ pertinent knowledge from prior time steps, which is crucial for precise climate forecasting over extended periods. Most of the literature reviewed focused on predicting future air temperature using historical data collected up to a given point in time. None of the methods performed a year-by-year analysis, i.e. whether data from a given year(s) is helpful to predict for future or past years. This type of detailed analysis could provide insights into the stability of fluctuations in weather patterns. Moreover, most of the approaches predicted one outcome, i.e. air or surface temperature. The application of advanced transformer models for tabular data remains unexplored in this field. This study, for the first time, utilizes four years worth of weather data collected in Kuwait. The key objectives are to (i) assess the role of previous year(s) in predicting the weather for a given year (past or future), and (ii) predict multi-output temperature outcomes using ML, sequential and tabular DL approaches. The next section describes the Kuwait climate data, data processing steps, feature extraction, and exploratory analysis conducted on this data

AUMET dataset

AUMET is a Weather Monitoring Station Weather Monitoring System (WMS) deployed at the research and innovation center of the American University of the Middle East (AUM)—Kuwait in 2019. AUMET is located at Latitude (North) 29° 9' 53.28" N, and Longitude (East) 48° 5' 53.61" E. The main component of this WMS is the GMX501 integrated weather sensor developed by GILL instruments. The entire WMS is situated on a 25-m-high elevated rooftop above ground level. The surface temperature sensor is placed ten centimeters above the rooftop floor. At the same time, all the other sensors in the integrated GMX501 instrument are mounted on a three-meter elevated mast above the rooftop surface. The WMS also includes a battery supplied by a 20W solar panel, a data logger, and a cellular antenna for remote access. Figures 1 and 2 shows the deployment and the general structure of the AUMET WMS.

The AUMET dataset contains weather and solar radiation data records with a 5-min resolution collected since August 2019. Each record of the collected dataset includes the measures of 14 solar and weather parameters. The dataset contains the minimum, maximum, and average values of some of these parameters along with a timestamp, as shown in Table 2; thus the total number of climate parameters is 33. The parameters related to average values are computed during the 5-min periods by the sensors, and they are not the average of minimum and maximum values. The parameter Windchill always gave a constant value. The Windchill values were kept as is during training the regression models because in future the sensor re-calibration may give meaningful values. It is expected that the regression models will disregard these constant values during training. The collected dataset has no duplicate values.

All the high-precision sensors in the integrated GMX501 unit are calibrated at the Gill Instruments® facility prior to deployment. For example, the internal 2-axis compass used for wind direction correction is fully calibrated before delivery. This ensures accurate magnetic north alignment without requiring field calibration. Furthermore, end-users can perform calibration checks by comparing GMX501 measurements against trusted reference instruments or by using a controlled environment, such as an environmental chamber or data from other sources. We have done this with a regional model to check the instrument functionality, and it was successful. The accuracy of the model which has been checked for ensuring measurement reliability is shown in Table 1. A crucial periodic practice is the routine checks that are performed on the AUMET WMS. And it is described as follows:

- Perform regular visual inspections for obstructions (e.g., dirt on rain gauges or radiation shields). Such inspection has been done on a quarterly basis for many years since the first installation.



Fig. 1. Installed WMS at AUM.

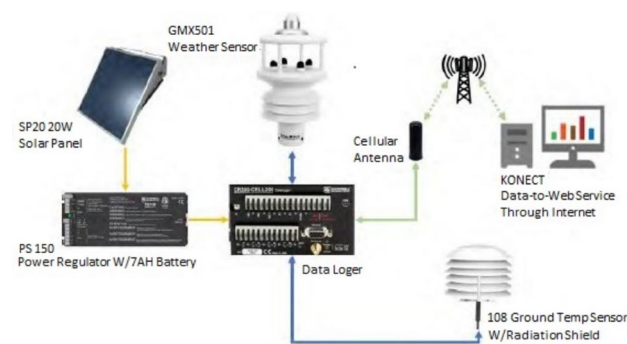


Fig. 2. AUMET WMS station diagram.

Parameter	Accuracy
Wind speed	$\pm 3\%$ (0.01 40 m/s), $\pm 5\%$ (40 60 m/s)
Wind direction	$\pm 3^\circ$ (0.01 40 m/s), $\pm 5^\circ$ (40 60 m/s)
Compass heading	$\pm 3^\circ$
Air temperature	$\pm 0.3^\circ \text{ C}$ at 20° C
Relative humidity	$\pm 2\%$ at 20° C (10 90% RH range)
Dew point	$\pm 0.3^\circ \text{ C}$ at 25° C
Barometric pressure	$\pm 0.5 \text{ hPa}$ at 25° C
Solar radiation	$\pm 2\%$ (using Hukseflux LPO2 pyranometer, ISO 9060 compliant)

Table 1. AUMET parameters accuracy.

- Compare field data with nearby certified meteorological stations to detect anomalies. Also, since measuring has been done for many years, there are no extreme shifts in trends of the measurements, which indicates the reliability of the sensor and data since installation.

In the year 2021, the solar panel malfunctioned and there was missing data in the months of January, March and May. A similar situation occurred in the year 2022 with missing data in the months of January, July, August and November. August 18 from 00:09:10–00:09:15 had two NaN rows. The year 2023 had only one missing row. The year 2024 had four NaN readings on March 10 from 00:09:00–00:09:15 and few other missing values in the month of January. The NaN readings refer to entire rows corresponding to a timestamp (or major parts of it) filled with NaN values due to the malfunction of one or more sensors. These rows were discarded from the datasets. Refer to Table 3 with detailed description on missing data per year. The year 2024 was a leap year, resulting in more data points than other years. Overall, the years 2021, 2022, 2023 and 2024 had 93073, 105063, 105119, and 105367 data points.

Data processing

For all the years, the dataset was first parsed into dependent and independent variables according to the research question. Six parameters related to the temperatures were selected as dependent variables, i.e. Air Temperature Average, Air Temperature Minimum, Air Temperature Maximum, Surface Temperature Average, Surface Temperature Minimum, and Surface Temperature Maximum. The rest of the parameters were considered as independent variables. The timestamp parameter is recorded in UTC+3, DD/MM/YYYY H:M:S format, which contains vital information about the day and time. The non-numeric nature of the timestamp excludes its utility in training regression models. Consequently, an additional feature extraction step is performed to extract hour, minute, day of the week, and month. The day of the week and month are numerically encoded from 1 to 7 and 1 to 12. Thus, the total number of features on which regressors are trained is 30, and they output six temperature related outcomes.

Exploratory analysis

An exploratory analysis is conducted on the data to identify any climatic idiosyncrasies or patterns. Figures 3 and 4 show the difference between the daily midday and midnight Air Temperature Average, and maximum and minimum Air Temperature Average temperature for each day of the year. While the overall distribution across these four years appears similar, each day's maximum and minimum temperature, and temperature during midday and midnight vary. We notice some spikes in the years 2021 and 2022 due to sudden changes in temperatures during some of the days. Therefore, simply using a previous year's data to predict for the next year is non-trivial. Due to missing data in the year 2021, a solid line appears in these plots. It is only a connecting line between two points representing missing data, and should not be confused with a valid temperature data.

Methods

It is important to distinguish between weather and climate time scales. Weather typically refers to short-term atmospheric conditions (minutes to days), while climate characterizes the statistical distribution of weather patterns over more extended periods (decades). Our dataset, sampled every 5 min over four years, lies between these scales: the fine temporal resolution captures weather-like variations, but the multi-year coverage enables assessment of climate-related stability. In this study, we frame the problem as high-frequency temperature prediction within a climate monitoring context, acknowledging that the models are trained on meteorological data but evaluated for their potential to support long-term climate resilience and planning in arid regions such as Kuwait.

In this study, we adopt a leave-1-year-out cross-validation strategy, where 3 years of data are used for training and the remaining year is held out for testing. The choice of a 1-year horizon ensures that the models are evaluated on complete seasonal cycles, capturing winter, summer, and transitional periods, which would not be possible with shorter hold-out intervals (e.g., weeks or months). From a practical standpoint, this setup reflects

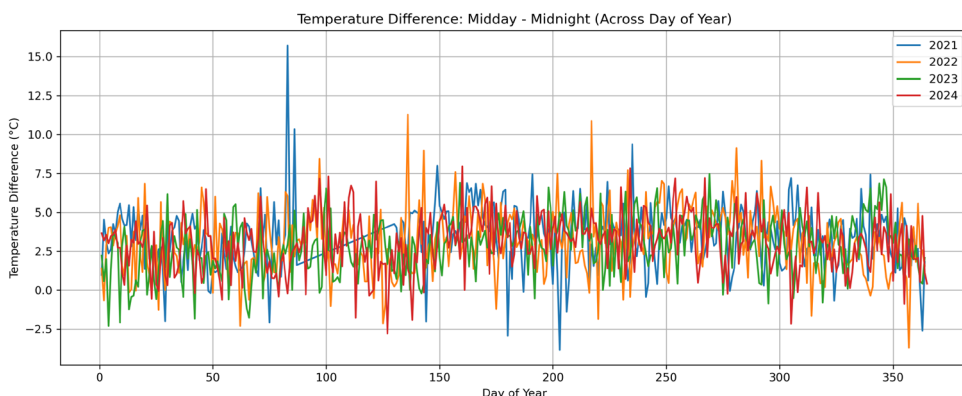


Fig. 3. Daily midday and midnight temperature difference for each day of the year.

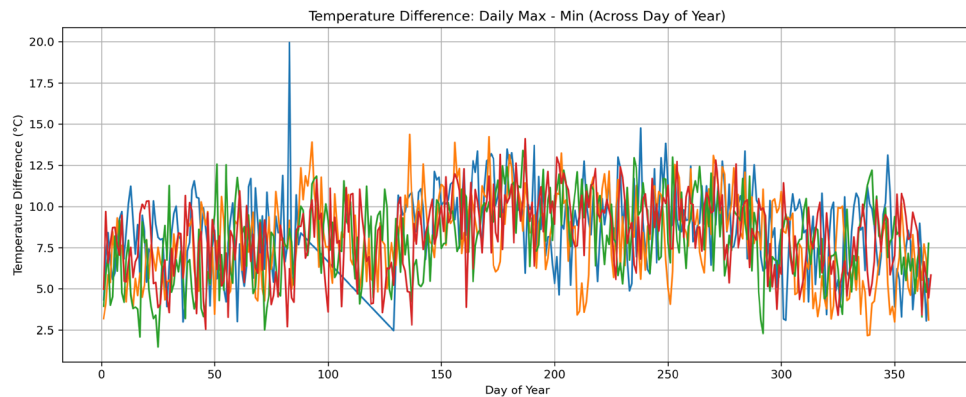


Fig. 4. Daily maximum and minimum temperature difference for each day of the year.

how predictive models may be deployed to forecast an upcoming year based on historical climate records, which is directly relevant for policy and planning in regions such as Kuwait that experience sharp seasonal variations.

Regression models

As the task is to predict six temperature variables (see Table 2), we formulate this as a multi-output regression problem. For performing a comprehensive analysis, we benchmark 10 regression models from ML, DL and deep sequential learning, including a naive baseline—as discussed below.

The naive baseline (BL) is basically the average of six temperature variables of a particular date and time from the 3 years, and uses it as a prediction for the fourth year. However, as per exploratory analysis, the prediction appears non-trivial. The other classical regressors used were Logistic Regression (LR), Random Forest (RF), Extra Trees (ET) and K-nearest neighbors (KNN). Two tabular DL regressors used in this analysis are the TabNet from Google⁴⁸ and FTTransformer from Yandex⁴⁹. TabNet employs a sequential attention technique to select features at every decision step, thereby enhancing interpretability and efficient learning. FTTransformer uses the attention mechanism to capture relationships between numerical and categorical tabular data. The attention mechanism dynamically assigns importance to certain features, allowing learning embeddings to better generalize. We employed three sequential DL based multi-output regressors based on LSTM, Temporal Convolutional Neural Networks (TCN) and Transformers, to capture temporal dependencies in the dataset to predict six temperature variables.

A 30-dimensional vector containing various climate variables is used to train various models, except the sequential models. In the sequential models, a 30-dimensional sequence of size 12 was created. This number was chosen to capture the last 60-min of variation (as data is collected every 5 min). Due to sequence formation, the last 12 timestamps could not be tested from each of the test sets. As a result, these last 12 rows (or timestamps) were not tested for each non-sequential regressor to ensure a fair comparison. All the DL models were run for 20 epochs. The data was scaled before being passed to each regressor (except RF and ET) and then re-scaled after the predictions were made to compute performance metrics. The ML models were trained with the default parameters. All the DL models were kept at a similar level of complexity for better comparison. The key parameters of these models are as follows:

- TabNet – latent dimensions = 64, number of decision steps = 5, attention relaxation parameter = 1.5, number of shared layers = 2, number of decision layer blocks = 2.
 - FTTransformer—token dimension = 64, feed-forward hidden layer size = 128, number of transformer layers = 4.
 - LSTM and TCN—hidden layer = 64, batch size = 64.
 - Transformer—batch size = 64, dimension of the input embeddings (including all hidden states) = 64, number of attention heads = 4, number of Transformer encoder/decoder layers = 2.
- Performance metrics** Three metrics were used to assess the performance of the multi-output prediction models of the temperature variables, namely the R^2 score, mean squared error (MSE), and mean absolute error (MAE). These metrics were averaged across the six output variables to give one value per metric.

Results

Leave-one-year-out cross validation

We trained the examined models in a leave-1-year-out (LOYO) manner, such that 3 years data is used for training and 1 year of data is used for testing. Table 4 shows the results with 10 regression models in a LOYO manner. Years 1, 2, 3, and 4 correspond to climate data from the years 2021, 2022, 2023, and 2024. The Test Year4 means the models were trained on data from Years 1, 2 and 3 and tested on Year4. Similarly, Test Year1 means that the models were trained on data from Years 2, 3, and 4, and tested on Year1, and so on. Table 5 shows the average, standard deviation and 95% confidence interval of R^2 , MSE and MAE for each regressor across all folds. The bold highlighted cells show the best performing regression models for a given test year. The results show that FTTransformer and LSTM models performed best in twofolds each. Among the traditional ML

#	Parameter	Unit	Description
1	Air_Density	kg/m ³	Density of the air
2	Air_Temp_Avg	° C	Average temperature of the air at three meters above the surface during 5 min
3	Air_Temp_Min	° C	Minimum temperature of the air at three meters above the surface during 5 min
4	Air_Temp_Max	° C	Maximum temperature of the air at three meters above the surface during 5 min
5	Corr_Wind_Dir_Avg	degree	Average corrected wind direction if the wind sensor is moving during 5 min
6	Corr_Wind_Dir_Min	degree	Minimum corrected wind direction if the wind sensor is moving during 5 min
7	Corr_Wind_Dir_Max	degree	Average corrected wind direction if the wind sensor is moving during 5 min
8	DPoint_Avg	° C	Average dew point temperature during 5 minutes. Dew point is the temperature at which air becomes saturated with moisture (water vapor)
9	DPoint_Min	° C	Minimum dew point temperature during 5 min.
10	DPoint_Max	° C	Maximum dew point temperature during 5 min
11	HeatIndex_C	° C	Represents how hot it feels to the human body when relative humidity is combined with the air temperature. It is also known as feels-like temperature in hot conditions
12	Pressure_Avg	hPa	Average air pressure
13	Pressure_Min	hPa	Minimum air pressure
14	Pressure_Max	hPa	Maximum air pressure
15	Relative_Wind_Dir_Avg	angleo	Average relative wind direction
16	Relative_Wind_Dir_Min	angleo	Minimum relative wind direction
17	Relative_Wind_Dir_Max	angleo	Maximum relative wind direction
18	Relative_Wind_Speed_Avg	m/s	Average wind speed
19	Relative_Wind_Speed_Min	m/s	Minimum wind speed
20	Relative_Wind_Speed_Max	m/s	Maximum wind speed
21	RH_Avg	pct	Average relative humidity
22	RH_Min	pct	Minimum relative humidity
23	RH_Max	pct	Maximum relative humidity
24	Solar_Radiation_Avg	w/m ²	Average amount of solar radiation in 5 min
25	Solar_Radiation_Min	w/m ²	Minimum amount of solar radiation in 5 min
26	Solar_Radiation_Max	w/m ²	Maximum amount of solar radiation in 5 min
27	Sunshine_Hr	H	The cumulative sunshine duration (in hours) during which direct solar irradiance is > 120 W/m ² recorded over a given day, and it is reported at 5-min intervals
28	Surface_Temp_Avg	° C	Average air temperature at 10 cm above the surface during 5 min
29	Surface_Temp_Min	° C	Minimum air temperature at 10 cm above the surface during 5 min
30	Surface_Temp_Max	° C	Maximum air temperature at 10 cm above the surface during 5 min
31	WetBulbTemp	° C	The Wet Bulb Temperature. it is the temperature a parcel of air would have if it were cooled to saturation
32	Windchill	–	999
33	Timestamp	UTC+3, DD/MM/YYYY H:M:S	Time of sensor reading

Table 2. Weather and solar parameters of the AUMET dataset.

models, RF regressor performed better, but its performance is worse than FTTransformer and LSTM. LSTM and FTTransformer performed equivalent across four-folds, RF and Transformer performed next. It is worth noting that LSTM is a simpler model than FTTransformer and faster to train, while achieving similar performance. FTTransformer does not model sequences but excels at learning robust inter-feature patterns across various space variables that may be more stable year-to-year than temporal patterns. The BL performed worse as expected, it shows that temperatures at a given time across years do not follow a simple fixed distribution and depends on variations of various climatic parameters. Thus, a naive baseline would be an oversimplification of this complex problem involving multiple climate variables.

Across the four annual test folds, both FTTransformer and LSTM achieved consistently high performance with mean $R^2 \approx 0.998$, $MSE \approx 0.13$, and $MAE \approx 0.2$. The 95% confidence intervals for these models were narrow, confirming the statistical robustness of their performance across unseen years (see Table 5). Violin plots for each performance metrics further illustrate this stability for both FTTransformer and LSTM (see Fig. 5a–c). In contrast, traditional machine learning models showed wider confidence intervals and broader violin plot distributions, reflecting greater variability and sensitivity to inter-annual differences. Together, the confidence intervals and violin plots demonstrate that deep learning models, especially FTTransformer, are not only accurate but also more stable across multiple yearly prediction tasks. Paired t-tests and Wilcoxon signed-rank tests between the two top-performing models (FTTransformer and LSTM) across the four folds showed no statistically significant differences in R^2 , MSE, or MAE ($p > 0.7$), indicating comparable predictive performance.

Year	Missing data period
2021	January 01, 00:00:00–January 01, 00:03:00
	March 29, 08:05:00–March 29, 08:10:00
	March 29, 08:20:00– March 29, 13:40:00
	March 29, 14:15:00– May 09, 11:25:00
	May 09, 18:20:00–May 10, 05:55:00
2022	January 01, 03:00:00 – January 01, 03:00:00
	July 21, 07:30:00 – July 21, 08:10:00
	August 18, 08:05:00 – August 18, 09:05:00
	August 18, 09:10:00 – August 18, 09:15:00 (two rows with NaN readings)
	August 18, 09:30:00 – August 18 09:30:00
	November 27, 09:45:00 – November 27, 12:15:00
2023	January 01, 03:00:00 – January 01, 03:00:00
2024	January 01, 03:00:00 – January 01, 06:00:00
	March 10, 00:09:00 – March 10, 00:09:15 (four rows with NaN readings)

Table 3. Missing data information in the AUMET dataset.

#	Regressors	Test Year4			Test Year3			Test Year2			Test Year1		
		R ²	MSE	MAE	R ²	MSE	MAE	R ²	MSE	MAE	R ²	MSE	MAE
1	BL	0.8951	10.4459	2.4716	0.9102	9.4265	2.4194	0.8893	9.7544	2.462	0.9045	9.5696	2.4181
2	LR	0.9645	3.5413	1.4638	0.963	3.8874	1.5937	0.9594	3.5849	1.508	0.9639	3.622	1.4984
3	RF	0.9981	0.195	0.2836	0.9986	0.1535	0.2561	0.9982	0.1641	0.2651	0.9982	0.1878	0.2743
4	ET	0.9972	0.2887	0.3549	0.9985	0.1635	0.2717	0.9981	0.1733	0.2817	0.9981	0.1997	0.2964
5	KNN	0.9656	3.425	1.4579	0.9619	4.0032	1.5372	0.9551	3.9577	1.5435	0.966	3.4106	1.4474
6	LSTM	0.9986	0.1468	0.2438	0.9989	0.1189	0.2168	0.9986	0.1294	0.2358	0.9986	0.1397	0.2392
7	TCN	0.998	0.1984	0.3142	0.9837	1.7073	0.3356	0.9976	0.2152	0.3104	0.9979	0.2138	0.3183
8	Transformer	0.9986	0.1451	0.248	0.9982	0.1893	0.2461	0.9985	0.1373	0.2363	0.9968	0.3213	0.2746
9	TabNet	0.9956	0.4462	0.4563	0.9979	0.2215	0.3409	0.9966	0.3041	0.3956	0.997	0.3081	0.4003
10	FTTransformer	0.9989	0.1121	0.2126	0.9986	0.1528	0.3193	0.9984	0.1473	0.241	0.9989	0.1097	0.2105

Table 4. Performance evaluation of various regressors on AUMET dataset (various metrics) for leave-1-year cross validation. The highlighted rows show the best performing regressor for a given testing year and performance metrics.

Due to the limited number of years (n = 4), the statistical tests should be interpreted with caution; however, they indicate robustness.

Impact of decreasing training data

In the experiments described in the previous “Leave-one-year-out cross validation” section, data from 3 years were used to train various regressors and the fourth year’s temperature variables were predicted. To understand the role of decreasing training data, we conducted two additional experiments. We chose Year4 as the current year (2024), and trained various regressors with data from past two years (Year3 and Year2), and past one year (Year3). The results from training the models on the past three years Year1, Year2 and Year3 are already available from Table 4. Table 6 shows the impact of decreasing the training data from the last three, two and one year to predict the six temperature variables for Year4. In terms of the R² score, the performance for all the regressors either improves or remain consistent (e.g. LSTM, Transformer) when more training data is used to train them. LSTM performed the best when it was trained on Year3 and tested on Year4, and its MSE was better than FTTransformer trained on Year3, Year2 and Year1. In terms of MAE, FTTransformer trained on Year3, Year2 and Year1 performed the best among all combinations.

One of the reasons for the degraded performance of LSTM when data from Year1 and Year2 were included in training is that the data from those years had more variations. It can be seen from Figs. 3 and 4 that Year1 and Year2 (i.e. 2021 and 2022) had many seasonal spikes and sudden temperature changes, while Year3 (2023) is more similar to Year4 (2024). As LSTM relies heavily on stable sequential patterns, misaligned training years’ data (including missing data in Year1) could have affected its generalization. On the other hand, FTTransformer learns time-independent interactions, making it less susceptible to sequence inconsistencies across years. Therefore, we could conclude that in this dataset, temporal dependencies may not be the dominant predictor as the distribution of climate data may vary across years (thus affecting LSTM’s performance). On the other hand, FTTransformer learned robust feature-wise relationships that are invariant across years.

Regressor	Metric	Mean	StdDev	95% CI Low	95% CI High
BL	R ²	0.8998	0.0094	0.8849	0.9147
	MSE	9.7991	0.4516	9.0805	10.5177
	MAE	2.4428	0.0280	2.3982	2.4874
LR	R ²	0.9627	0.0023	0.9591	0.9663
	MSE	3.6589	0.1559	3.4109	3.9069
	MAE	1.5160	0.0552	1.4282	1.6038
RF	R ²	0.9983	0.0002	0.9979	0.9986
	MSE	0.1751	0.0195	0.1440	0.2062
	MAE	0.2698	0.0118	0.2509	0.2886
ET	R ²	0.9980	0.0006	0.9971	0.9989
	MSE	0.2063	0.0570	0.1156	0.2970
	MAE	0.3012	0.0372	0.2419	0.3604
KNN	R ²	0.9622	0.0050	0.9541	0.9702
	MSE	3.6991	0.3254	3.1813	4.2170
	MAE	1.4965	0.0509	1.4155	1.5775
LSTM	R ²	0.9987	0.0001	0.9984	0.9989
	MSE	0.1337	0.0122	0.1143	0.1531
	MAE	0.2339	0.0119	0.2150	0.2528
TCN	R ²	0.9943	0.0071	0.9831	1.0055
	MSE	0.5837	0.7491	− 0.6083	1.7757
	MAE	0.3196	0.0111	0.3019	0.3373
Transformer	R ²	0.9980	0.0008	0.9967	0.9994
	MSE	0.1982	0.0852	0.0627	0.3338
	MAE	0.2512	0.0164	0.2252	0.2773
TabNet	R ²	0.9968	0.0010	0.9953	0.9983
	MSE	0.3200	0.0931	0.1718	0.4682
	MAE	0.3983	0.0472	0.3232	0.4733
FTTransformer	R ²	0.9987	0.0002	0.9983	0.9991
	MSE	0.1305	0.0227	0.0943	0.1667
	MAE	0.2459	0.0509	0.1649	0.3268

Table 5. Performance summary of regression models with mean, standard deviation, and 95% confidence intervals for different regressors across all years on the AUMET dataset.

Monthly predictions

We plotted the monthly predictions for LSTM and FTTransformer for testing Year4, when the model is trained on climatic data from Year1, Year2, and Year3. Figure 6 shows that FTTransformer performs best for all months in terms of R² score, MSE and MAE, with the exception of April (for MSE) and August (for MAE). The overall better performance of FTTransformer in comparison to the next best model of LSTM can be explained by its superior performance across each month with varying temperatures. This also implies that FTTransformer robustly model the non-linear relationships between different climate variables throughout the years in predicting multi-output temperature outcomes.

Model explainability

In order to explain the best performing model, FTTransformer, we conducted two analyses—SHAP⁵⁰ and permutation importance⁵¹. We present the results of testing FTTransformer on Year4, where the model was trained on the preceding 3 years. In this case, SHAP analysis can be used to interpret the prediction of an FTTransformer by explaining the contribution of each feature to the regression task. In contrast, permutation importance measures feature importance by randomly shuffling the values and assessing their impact on the regression performance. SHAP can indicate the impact on performance by changing feature values, whereas permutation importance cannot. SHAP can handle complex interrelationships between features; however, it can be highly computationally demanding.

SHAP analysis We conducted the SHAP analysis on 500 samples from the test set due to time constraints. Figure 7 shows the SHAP summary plot highlighting the top weather features that impact the model outcomes. It can be observed that higher and lower values for DPoint_C_Avg push the model prediction up and down; this feature appears to be the most important. The CorrWindDir_deg_Min has a mixed impact, with some values increasing (red) and others decreasing (blue) predictions. Its impact further depends on other features (e.g. DPoint_C_Avg); thus it has a lower effect than DPoint_C_Avg. AirDensity_Kgpm3 has a lower influence than DPoint_C_Avg and DPoint_C_Max, indicating a negative trend with higher density may weakly reduce

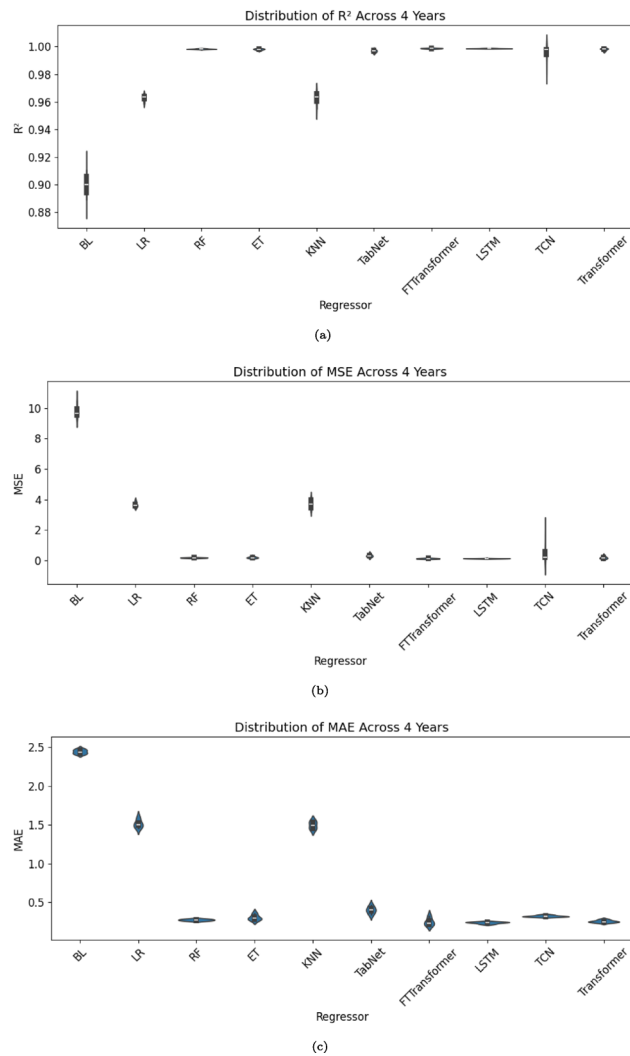


Fig. 5. Violin plots for (a) R^2 score (b) MSE (c) MAE, for each regressor across four years.

Regressor	Training: Year3			Training: Year3+Year2			Training: Year3+Year2+Year1		
	Testing: Year4			Testing: Year4			Testing: Year4		
	R^2	MSE	MAE	R^2	MSE	MAE	R^2	MSE	MAE
BL	0.8600	13.9401	2.9232	0.8836	11.5912	2.6355	0.8951	10.4459	2.4716
LR	0.7275	27.0839	1.4772	0.9627	3.7128	1.5031	0.9645	3.5413	1.4638
RF	0.9981	0.1959	0.3058	0.9977	0.2308	0.3047	0.9981	0.1950	0.2836
ET	0.9934	0.6561	0.4994	0.9945	0.5503	0.4621	0.9972	0.2887	0.3549
KNN	0.9498	5.0009	1.7447	0.9581	4.1696	1.6009	0.9656	3.4250	1.4579
LSTM	0.9991	0.0944	0.2217	0.9989	0.1161	0.2316	0.9986	0.1468	0.2438
TCN	0.9804	1.9477	0.3902	0.9811	1.9042	0.3672	0.9980	0.1984	0.3142
Transformer	0.9986	0.1392	0.2841	0.9987	0.1371	0.2445	0.9986	0.1451	0.2480
TabNet	0.9839	1.6037	0.9050	0.9945	0.5558	0.5330	0.9956	0.4462	0.4563
FTTransformer	0.9971	0.2866	0.3997	0.9986	0.1449	0.2673	0.9989	0.1121	0.2126

Table 6. Impact of increasing training data for predicting six temperature outcomes for Year 4 using different regressors.

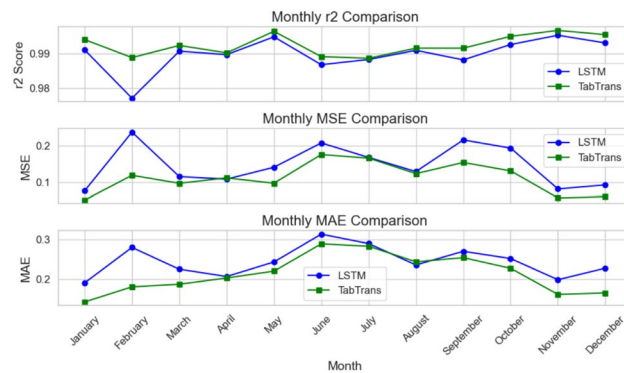


Fig. 6. Monthly predictions for LSTM and FTTransformer for Year4 when the models are trained on data from Year1, Year2 and Year3.

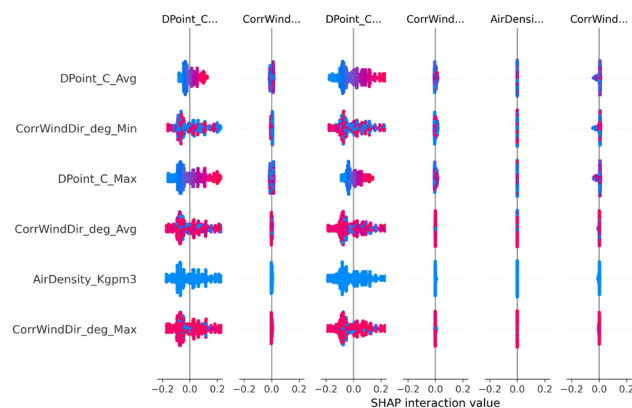


Fig. 7. SHAP summary plot showing the impact of top weather features on model predictions. Each dot represents a sample ($n = 500$), with feature values color-coded (red = high, blue = low). Horizontal axis shows SHAP values (higher = positive prediction impact); vertical axis lists features by importance (top to bottom). Dew point (DPoint_C_Avg) dominates predictions, while wind direction (CorrWindDir_deg) exhibits context-dependent effects.

prediction. In summary, features related to dew point turn out to be the most important, followed by wind direction (which depends on dew point). At the same time, air density is comparatively less critical for this model's performance.

Permutation importance We also employed the permutation importance technique⁵¹ to explain features, which is model agnostic and can be used for global explainability. In this approach, for each feature, the values are randomly shuffled to break the relationship with the target and measure the drop in model performance (here, averaged MSE across all six outputs). A significant performance drop indicates that a feature is important. Figure 8 shows the feature importance for the FTTransformer model. It can be observed that relative humidity and dew point features are the most important, meaning that if their values are shuffled, the model's predictions degrade significantly.

The features identified by permutation importance and SHAP analysis may differ due to the distinct criteria employed by each method to assess feature significance. While SHAP measures the contribution of a feature to a single prediction, permutation importance measures the impact on model performance when a feature is corrupted. The scope of SHAP is local and captures features interaction, whereas permutation importance's scope is global and treats features independently. From both analyses, it is clear that dew point features are the most important and humidity, wind direction and air density are also useful. This further justifies the necessity of collecting different climate variables to ensure accurate multi-output temperature predictions.

Impact and limitations

Our results showed that high resolution and multi-output temperature parameters (six in our case) in Kuwait from 2021 to 2024 can be predicted with high reliability from other climate variables in a manner that generalizes across 4 years. The FTTransformer and LSTM regressor emerged as the best choice among the ten different regressors including a baseline, ML, DL and sequential models. This result has implications for sensor fault analysis, missing data imputation and understanding drivers for variations in temperatures in microclimate situations. The high fidelity results lead to other critical scenarios, such as predicting temperatures without direct

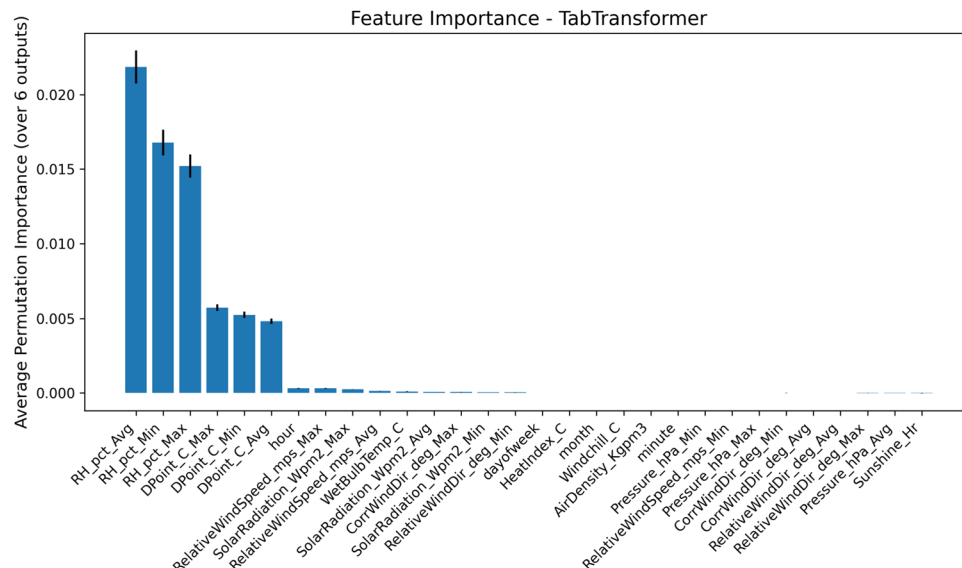


Fig. 8. Average permutation importance of input features for the FTTransformer model across six target variables. The bar heights represent the mean decrease in model performance when each feature is randomly shuffled, with error bars indicating standard deviation. Relative Humidity and Dew Point variables exhibit the highest importance, indicating strong global influence on the model's predictions.

temperature sensors in situations where these sensors may fail or cannot be calibrated. The results are also valuable for reducing sensor redundancy, missing data reconstruction, quality control and sensor validation. The multi-output LOYO CV results are non-trivial and demonstrate the stability of the weather in the Kuwait region during the study period. This approach can be used for deployments where only certain sensors are feasible. The multi-output modeling of six temperature variables at a high frequency (every 5 min) is valuable for urban heat studies, building energy models, agricultural and environmental control systems. Moreover, the LOYO CV approach is a sound experimental design that provides a reproducible framework and benchmark for others using similar environmental/ecological setups.

The overall four years span of the AUMET data is insufficient to infer climate changes due to its slow progression. The 5-min sampling rate of each data point could also overwhelm sequential models with a large amount of training data, thus hindering their generalization capabilities. We used a sequence length of 12 samples, which may be insufficient to capture long-term temporal dependencies, such as diurnal patterns, sudden or anomalous climate change across days and delayed effects of solar radiation and humidity. Therefore, sequence length serves as a hyperparameter in this context and may be adjusted to inform decisions. The AUMET dataset has several missing values; model performance could be improved by adequately imputing these values. The models were trained on 30 climate variables, including features extracted from the time of the day. The regression models could be improved by incorporating additional contextual information, such as geography, air quality, cloud cover, wind dynamics, and elevation. Techniques based on domain adaptation and co-variate shift detection could help in modeling divergent years (e.g., Years 1 and 2 in our case). The DL models were trained for only 20 epochs, increasing the number of training epochs could further improve model performance. The overall study findings are limited by Kuwait specific regional data that may not be generalizable to other geographical regions.

Conclusions and future work

We conducted a comprehensive investigation of predicting multi-output air and surface temperatures in Kuwait for the years 2021 to 2024 based on various climate variables. We introduced a LOYO CV framework to establish relationship between multiple years' climate data in predicting six temperature outcomes simultaneously. This study demonstrates the efficacy of advanced multi-output deep learning models for high-frequency temperature forecasting in arid environments. Among the ten regression models, FTTransformer and LSTM achieved an average R^2 of 0.998, with MSE of 0.13 and MAE of 0.24 across unseen years. Overall, FTTransformer emerged as the most reliable regressor, with LSTM and RF as other simpler options but with lower performance. LSTM struggles to generalize when either past data contained anomalous or missing values. These results indicate that even fine-grained, 5-min interval temperatures can be reliably predicted from other climate variables. Model explainability analysis showed the importance of collecting various climate variables in making accurate multi-output temperature predictions.

While our results demonstrate the ability of multi-output deep learning models to predict high-frequency temperature variations, we emphasize that this work is positioned within a climate monitoring context rather than operational weather forecasting. The 5-min resolution data allow us to capture weather-scale variability, but the multi-year scope highlights inter-annual stability, making the findings most relevant for long-term climate resilience planning in Kuwait. These findings have practical implications for ecotourism planning and related

sectors in Kuwait. Accurate temperature forecasts at daily and seasonal horizons can help optimize outdoor recreational activities, inform safe scheduling of desert tours, and guide infrastructure planning for tourist comfort during peak heat months. More broadly, robust short- to medium-term forecasts enhance preparedness for extreme temperature events, which are critical for sustaining ecotourism in arid regions vulnerable to climate extremes.

In the future, we will perform time series anomaly detection analysis to identify abnormal time variations and the reasons leading to them. The robust performance of FTTransformer across several years can be used to impute missing climate data to assess whether it improves the overall regression performance. Imputing missing values could improve the performance of sequential models (e.g., LSTM, TCN, Transformer) because some of the rows were removed due to missing data that could have led to incorrect sequences. A near term objective is to utilize this data for forecasting temperatures over both short- and long-term horizons and at various time resolutions, as well as assessing the stability of forecasting the climate in the Kuwait region using relevant metrics, such as Theil's U2 and the Nash-Sutcliffe Efficiency. Lastly, the data from different weather stations across Kuwait could be used to cross-validate each other and build reliable climate modeling techniques.

Data availability

The data that support the findings of this study are available from the American University of the Middle East, Kuwait but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available. Data are however available from the authors upon reasonable request and with permission of the American University of the Middle East, Kuwait.

Code availability

The general code will be released upon request to support similar analysis.

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Author contributions

S.K. formulated the temperature prediction as a multi-output regression problem. R.A. collected and curated the data and provided technical support. S.K. performed and evaluated the experiments, and R.A. validated them. S.K. wrote the first draft with support from R.A. and both reviewed the paper together.

Declarations

Competing interests

The authors declare no competing interests.

Consent for publication

Both the authors consent for publications.

Additional information

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